KNECK OFFICIAL

Mathematics — Paper 2

121/2 • ORIGINAL REVISION SET 01 • 2 hours 30 minutes • 100 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer ALL of Section A and ANY FIVE questions in Section B. Give non-exact results to 4 significant figures unless stated otherwise.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SECTION A

All 16 questions: 50 marks.

1. Solve 2^(x+1) = 32. [3 marks]

2. Evaluate log10(1000) + log10(0.01). [3 marks]

3. Find the inverse of f(x) = (3x - 2)/4. [3 marks]

4. Differentiate $y = 3x^3 - 4x^2 + 5x - 7$. [3 marks]

5. Evaluate the integral of (4x + 3) from x = 1 to x = 3. [3 marks]

6. Find the determinant and inverse of matrix [[2,1],[3,2]]. [3 marks]

7. Solve $2\sin\theta = 1$ for $0° \leq \theta \leq 360°$. [3 marks]

8. A sequence has nth term 5n - 2. Find the first term, the 12th term and the common difference. [3 marks]

9. Expand (1 + 2x)^4 up to and including the x^2 term. [3 marks]

10. The radius of a sphere is measured as 7.0 cm to the nearest 0.1 cm. State radius bounds and calculate the upper volume bound using π = 3.142. [3 marks]

11. A sum of KSh 20,000 earns 10% compound interest annually. Find its value after 2 years. [3 marks]

12. If y varies inversely as x^2 and y = 8 when x = 3, find y when x = 6. [3 marks]

13. Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$. [3 marks]

14. A point P(2,-3) is reflected in y = x and then translated by (4,1). Find the final image. [3 marks]

15. Solve $x^2 - 4x - 1 = 0$, giving answers in surd form. [4 marks]

16. Find the equation of the tangent to $y = x^2 + 1$ at x = 2. [4 marks]

SECTION B

Answer any FIVE; 10 marks each.

17. A quadratic curve is $y = x^2 - 4x + 3$. (a) Complete a value table for x = -1,0,1,2,3,4,5. (3) (b) Draw the graph on graph paper. (3) (c) State roots and turning point. (4) [10 marks]

18. A machine costs KSh 800,000 and depreciates by 12% annually on reducing balance. (a) Find value after three years. (4) (b) Find total loss. (2) (c) Find the least whole number of years for value to fall below KSh 400,000. (4) [10 marks]

19. A and B are at latitude 30°N, longitudes 20°E and 80°E. Take Earth radius 6,370 km and π = 3.142. (a) Find the radius of their parallel. (3) (b) Find the distance along the parallel. (4) (c) Find their difference in local solar time. (3) [10 marks]

20. A particle displacement is $s = t^3 - 6t^2 + 9t$ metres. (a) Find velocity and acceleration functions. (4) (b) Find times of rest. (3) (c) Find acceleration at t = 4. (3) [10 marks]

21. A bag holds 4 red and 6 blue counters. Two are drawn without replacement. (a) Draw a labelled tree. (4) (b) Find probability of same colour. (3) (c) Find probability of exactly one red. (3) [10 marks]

22. A farmer has at most 40 ha for maize x and beans y. Maize needs 3 labour-days/ha, beans 2; available labour 100 days. At least 10 ha beans must be grown. Profit is KSh 30,000/ha maize and KSh 20,000/ha beans. (a) Write inequalities. (3) (b) Draw feasible region. (3) (c) Find maximum profit and state whether the optimum is unique. (4) [10 marks]

23. The first term of a geometric progression is 3 and common ratio 2. (a) Find the 8th term. (3) (b) Sum first 8 terms. (3) (c) An arithmetic progression has first term 3 and difference 2. Find its first 8-term sum and compare. (4) [10 marks]

24. Triangle ABC has AB=8 cm, AC=11 cm and angle BAC=60°. (a) Find BC. (4) (b) Find area. (3) (c) Find angle ABC. (3) [10 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SECTION A

1. (3 marks) 32 = 2^5; x+1 = 5; x = 4.

2. (3 marks) 3 + (-2) = 1; one mark each logarithm and sum.

3. (3 marks) y=(3x-2)/4; x=(4y+2)/3; inverse f^-1(x)=(4x+2)/3.

4. (3 marks) dy/dx = $9x^2$ - 8x + 5; one per nonconstant term.

5. (3 marks) Antiderivative $2x^2+3x$; values 27 and 5; difference 22.

6. (3 marks) Determinant 1; inverse [[2,-1],[-3,2]]. 1 determinant and 2 inverse.

7. (3 marks) sinθ = 1/2; θ = 30°,150°; one per stage.

8. (3 marks) 3; 58; 5. One mark each.

9. (3 marks) 1 + 8x + $24x^2$. Binomial coefficients 1,4,6; correct substitution.

10. (3 marks) 6.95 ≤ r < 7.05; upper bound $(4/3)\times3.142\times7.05^3 \approx 1467.95$ cm^3. 2 bounds, 1 correct substitution/result.

11. (3 marks) $20000(1.10)^2$ = KSh 24,200; formula, substitution, answer.

12. (3 marks) $y=k/x^2$; k=72; y=2.

13. (3 marks) $(x-3)^2+(y+2)^2=25$; centre (3,-2); radius 5.

14. (3 marks) Reflection (-3,2); add (4,1); image (1,3).

15. (4 marks) Quadratic formula or completing square; $x=2\pm\sqrt{5}$. Method 2, each root 1.

16. (4 marks) Point (2,5); gradient 2x=4; y-5=4(x-2); y=4x-3.

SECTION B

17. (10 marks) Values 8,3,0,-1,0,3,8 (3). Axes/scale 1, plotted points 1, smooth curve 1. Roots 1,3 (2); vertex (2,-1) (2).

18. (10 marks) $800000(0.88)^3$ = 545177.60 (4); loss 254822.40 (2). (0.88)^n<0.5; n>log(0.5)/log(0.88)=5.422; minimum 6 years (4).

19. (10 marks) r=6370cos30°=5516.58 km (3). Arc=(60/360)×2×3.142×r≈5777.70 km (4). 60×4=240 min=4 h; B is ahead (3).

20. (10 marks) $v=3t^2-12t+9$; a=6t-12 (4). 3(t-1)(t-3)=0; t=1,3 s (3). a(4)=12 m/s^2 (3).

21. (10 marks) First 4/10,6/10; second 3/9,6/9 after R and 4/9,5/9 after B (4). Same=12/90+30/90=7/15 (3). Exactly one red=24/90+24/90=8/15 (3).

22. (10 marks) $x \geq 0, y \geq 10, x+y \leq 40, 3x+2y \leq 100$ (3). Vertices (0,10),(0,40),(20,20),(80/3,10) (3). Profit=10000(3x+2y), maximum 1,000,000; any point on feasible segment from (20,20) to (80/3,10) is optimal, hence not unique (4).

23. (10 marks) T8=3×2^7=384 (3). S8=3(2^8-1)=765 (3). AP sum=8/2(6+14)=80; GP exceeds AP by 685 (4).

24. (10 marks) $BC^2=64+121-176\cos 60=97$; $BC=\sqrt{97}\approx 9.849$ cm (4). Area=1/2×8×11sin60=38.11 cm² (3). cosB=(64+97-121)/(16√97); B≈75.30° (3).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf